Computer Peripherals: Assignment Answers

Printers, Scanners, Webcams and Connection Interfaces

Question 1

Create a table listing 3 types of printers, 2 types of scanners, and 2 types of webcams, and for each, write one real-life use case (for example: 'Inkjet printer – printing college assignments').

Device	Type	Main characteristic	Real-life use case
Printer	Dot-matrix printer	Impact printer; inexpensive, low cost per page, but noisy	Printing bills and bulk text where low running cost matters
	Inkjet printer	Non-impact printer; high-quality print (about 300 dpi as standard)	Printing college assignments and photographs at home
	Laser printer	Non-impact printer; 600 to 1200 dpi, high-quality text and graphics	Printing office documents and reports
Scanner	Flatbed scanner	Also called a desktop scanner; the document is placed face down on a glass plate	Scanning certificates, photographs and book pages
	Sheet-fed scanner	Pages are fed into the scanner; cannot scan bound documents	Scanning a stack of loose application forms
Webcam	Built-in webcam	Camera fitted into the laptop screen frame	Attending online classes and meetings on a laptop
	External USB webcam	Separate camera connected to the computer by a USB cable	Video conferencing from a desktop computer

Question 2

Draw a labeled diagram showing how a typical home printer connects to a laptop or PC, including the connection interface used (USB, Wi-Fi, or Bluetooth).

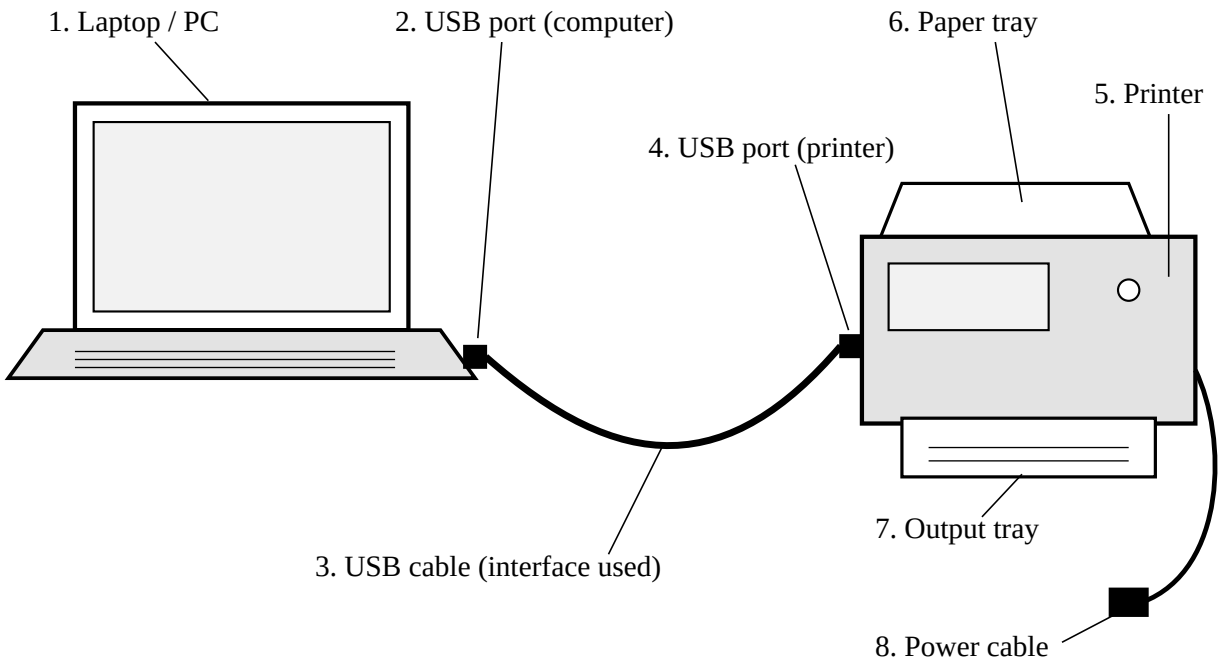

Figure 1: Home printer connected to a laptop or PC.

No.	Part	Function
1	Laptop / PC	Sends the document to be printed
2, 4	USB ports	Sockets on the computer and printer where the cable plugs in
3	USB cable	Carries data from the computer to the printer (Type-A end to Type-B end)
5, 6, 7	Printer, paper tray, output tray	Prints the job; paper is loaded in the tray and comes out at the output tray
8	Power cable	Supplies mains power to the printer

Interface used: USB. The computer must also have the printer driver installed. The printer can alternatively be connected over Wi-Fi or Bluetooth if it supports them.

Question 3

List three different connection interfaces (such as USB, Wi-Fi, Bluetooth) used by peripherals like printers, scanners, and webcams, and for each, name one advantage and one disadvantage. (Hint: Think about speed, convenience, and compatibility.)

Interface	Advantage	Disadvantage
USB	Most widely used connection point for data transfer; it replaced the serial, parallel and PS/2 ports, so it is compatible with most devices, and it also supplies +5 V power.	Needs a physical cable, so the device must be placed close to the computer.
Wi-Fi	Convenient wireless access from almost any location; range of up to about 100 feet outdoors and 400 feet indoors.	Speed often fluctuates, is generally slower than wired connections, and the network can be affected by interference and security weaknesses.
Bluetooth	Cable-free connection that can transmit data through walls; newer versions are backward compatible with older ones.	Limited to a short distance (about 10 feet) and has a lower data rate than USB.

Summary

In terms of **speed**, USB is the most dependable because it is wired. For **convenience**, Wi-Fi and Bluetooth remove the cable, with Wi-Fi covering the larger area. For **compatibility**, USB is found on almost every computer and peripheral, while Bluetooth depends on both devices supporting a compatible version.

Question 4

Pick any one device you use at home or college (printer, scanner, or webcam). Write step-by-step instructions for how you would connect it to your computer, including what cable or wireless method you would use and any setup steps required.

Device chosen: printer. **Connection method:** USB cable.

Requirements

Printer, power cable, USB cable (Type-A to Type-B), driver software (supplied CD or manufacturer's website), paper and ink or toner.

Procedure

1. Check the system requirements in the printer's user manual to confirm that the computer meets the minimum hardware and software requirements.
2. Place the printer near the computer and plug in its power cable.
3. Connect the printer to the computer with the USB cable, then insert the software CD supplied with the printer.
4. Open the CD drive, double-click the Setup file and follow the on-screen instructions. Alternatively, Windows detects a USB printer automatically and installs the required driver.
5. If the installation fails, download the updated driver from the printer manufacturer's website (for example, the Canon website for a Canon printer) and install it.
6. Print a test page to confirm that the printer is working correctly.

Wireless alternative

Connect the printer to the same Wi-Fi network as the computer and add it from the computer's printer settings.